

Woocheol Shin

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Research Interest

Education

KAIST

M.S. Graduate School of Data Science

Feb. 2025 ~ Present

UNIST

B.S. Computer Science and Engineering

Feb. 2019 ~ Feb. 2025

Research Experience

○ System Intelligence Lab, KAIST

Undergraduate Research Intern

Advisor : Prof. Jinkyoo Park(jinkyoo.park@kaist.ac.kr)

Oct. 2024 ~ Feb. 2025

○ Data Mining Lab, UNIST

Undergraduate Researcher, Seminar Organizer

Advisor : Prof. Junghoon Kim (junghoon.kim@unist.ac.kr)

Focus : Network stability, Cohesive subgraph discovery

Dec. 2022 ~ Jun. 2024

Research Papers

○ Diffusion Fine-Tuning via Reparameterized Policy Gradient of the Soft Q-Function

Hyeongyu Kang, Jaewoo Lee, Woocheol Shin, Kiyoungh Om, Jinkyoo Park

ICLR 2026

○ Diffusion Alignment as Variational Expectation-Maximization

Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin,

Taeyoung Yun, Kiyoungh Om, Jinkyoo Park

ICLR 2026

○ Urban Traffic Network Layout Optimization with Guided Discrete Diffusion Models

Taeyoung Yun, Inhyuck Song, Woocheol Shin, Yujin Shin, Sungpil Woo, Sunghwan Lim, Jinkyoo Park

WSDM 2026

○ Beyond Trivial Edges: A Fractional Approach to Cohesive Subgraph Detection in Hypergraphs

Hyewon Kim, Woocheol Shin, Dahee Kim, Junghoon Kim, Sungsu Lim, Hyunji Jeong

Knowledge-Based Systems 2025

○ Complex Network Stability: Exploring Comprehensive Anchored Edge s-Core

Taejoon Han, Hyewon Kim, Seongsik Hwang, Woocheol Shin, Junghoon Kim, Jaehyun Bhang, Hyeonggi Hong

Korea Database Conference (WiP Track) 2024

Awards and Achievements

○ ISE & GSDS Research Day Poster Competition, KAIST

Excellence Award

For the poster *Diffusion Finetuning via Reparameterized Policy Gradient of the Soft Q-function*

Nov. 2025

○ Dean's List, UNIST CSE

Awarded to top 4 students out of 224

Sep. 2024

○ Superior semester grades, UNIST

Spring 2023 (4.3/4.3), Fall 2023 (4.24/4.3), Spring 2024 (4.3/4.3)

Scholarship

- KAIST Scholarship, KAIST Whole semesters
- Academic Performance Scholarship, UNIST Whole semesters

Patent

- Data Analysis System and Method for Detecting Cohesive Subgraph in Hypergraphs Korean Patent Application, Aug. 2025
 - Application No. 10-2025-0107601
 - Applicants: UNIST; Industry-Academic Cooperation Foundation, Kongju National University
 - Inventors: Junghoon Kim, Hyewon Kim, Woocheol Shin, Dahee Kim, Hyunji Jeong

Teaching Experience

- Teaching Assistant, KAIST Sep. 2025 – Dec. 2025
 - IE437: Data-Driven Decision Making and Control

Project

- AICP, UNIST Apr. 2023 ~ Dec. 2023
 - UNIST AI Challengers Program
 - Topic : Bipartite Cohesive Subgraph Discovery in Networks
- Creative Independent Project, UNIST AIGS Oct. 2023 ~ May. 2024
 - Topic : A Budget Minimisation to Reinforcing Social Engagement
- UIRP, UNIST Mar. 2024 ~ May. 2024
 - UNIST Undergraduate Interdisciplinary Research Project
 - Topic : Generalised Anchoring Methods for Maximising User Engagement in Complex Social Networks

Industrial Projects

- Collaborative Project with ETRI Feb. 2025 ~ Present
 - Topic : Reinforcement Learning-based Job Scheduling Policy Learning

Other Experiences

- NAVER CONNECT 1784 School Revolution Program, NAVER Dec. 2022 ~ Feb. 2023
 - Teaching AI, SW, and Autonomous Vehicles
- ARAMCO Coding school, ARAMCO Aug. 2022 ~ Nov. 2022
 - Teaching Programming, IOS app development
- Education donation activities by university students, KOFAC May. 2019 ~ Jul. 2019
 - Teaching Programming

References

- Assistant Professor, Junghoon Kim
 - Computer Science and Engineering
 - UNIST
 - Email: junghoon.kim@unist.ac.kr
- Associate Professor, Jinkyoo Park
 - Department of Industrial and Systems Engineering
 - Associate Professor (Adjunct), Graduate School of Data Science
 - Associate Professor (Adjunct), Graduate School of AI
 - KAIST
 - Email: jinkyoo.park@kaist.ac.kr

